DOC-ANALOG-003

1. Validation Objectives and Rationale for the Simulation

The purpose of this document is to validate the small-signal transfer function $G_{vd}(s)$—calculated analytically in DOC-ANALOG-002—by running the admittance matrix model in the LTspice circuit simulator.

To perform a continuous *.ac* analysis in the frequency domain without the nonlinear interference caused by 100 kHz switching, the Vorperian Averaged Switch Model (PWM Switch Model in CCM) has been implemented. The switching branch is replaced by the behavior-dependent source *bv*, which linearizes the instantaneous duty cycle at the static operating point:

$$v_{sw}(t) = D \cdot V_{in} + V_{in} \cdot \hat{d}(t)$$

Where

$$D = \frac{V_{out}}{V_{in}} = \frac{5.0\ V}{24.0\ V} \approx 0.20833$$

2. Netlist Topology and Circuit Capture

The circuit implemented in LTspice integrates the V_{in} bus power supply, the small-signal excitation source V_d with a unit disturbance (AC = 1 V), the switching behavioral block B_1, and the power *LC* filter with the actual parasitic impedances ($DCR \equiv r_L = 25$ mΩ, $ESR \equiv r_C = 25$ mΩ).

2.1. Formal Netlist Executed

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* Circuit: Buck Converter Small-Signal Open-Loop Plant (PWMVM)
B1 N001 0 V=0.20833 * V(in_node) + 24 * V(duty_ac)
L1 N001 N002 68µ Rser=25m
C1 N002 0 220µ Rser=25m
R1 N002 0 2.5
V1 in_node 0 24
V2 duty_ac 0 AC 1
.ac dec 100 10 1Meg
.backanno
.end
```

2.2. Screenshot of the schematic in LTspice

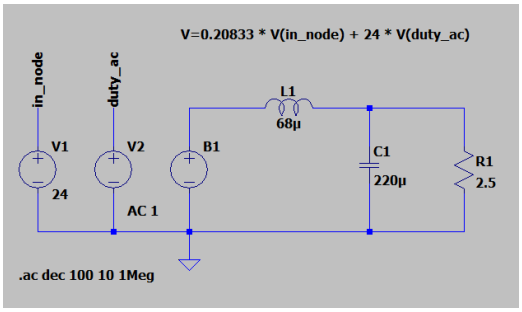


Figure 1: Simulation of the circuit in LTspice using the B_1 behavioral source for open-loop small-signal analysis.

3. Results Measured in LTspice and Bode Plot

When the *.ac dec 100 10 1Meg* command is applied, the frequency spectrum is swept from 10 Hz to 1 MHz with a resolution of 100 points per decade. The voltage response is measured at output node N002 (V_{out}).

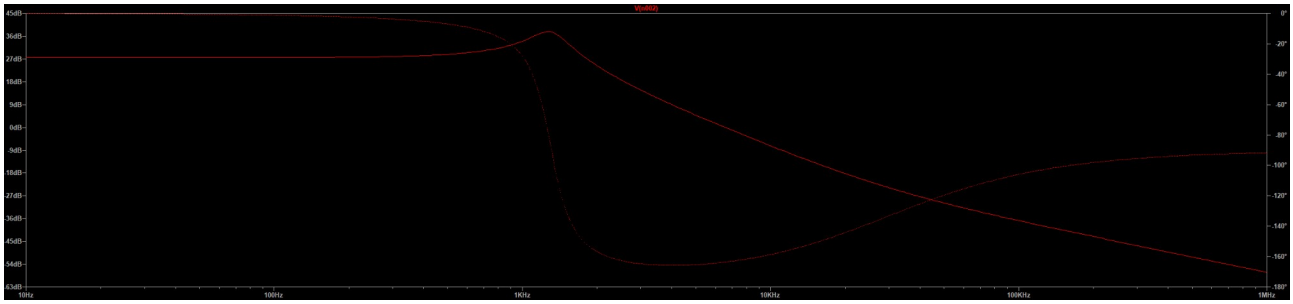


Figure 2: Open-loop Bode plot generated in LTspice. The DC gain, the resonance peak of the LC filter, and the phase drop are shown.

3.1. Comparison Table: Theoretical Formulation vs. Actual Measurement in SPICE

Dynamic Parameter	Theoretical Calculus (DOC-ANALOG-002)	Actual Measurement in LTspice	Relative Error (%)	Conclusión Técnica
DC Gain (10 Hz)	23.762 V (27.518 dB)	27.518 dB (Phase -0.117º)	0.00 %	Exact match in attenuation due to *DCR* (r_L).
Resonant Frequency (f_0)	1,301.23 Hz	1,300.00 Hz	0.09 %	Confirmation of the filter's natural resonance frequency *LC*.
Resonance Magnitude (f_0)	+37.700 dB	+37.675 dB	0.06 %	Real resonance peak Q = 3.22 (+10.16 dB above the DC baseline).
Resonance Phase (f_0)	-90.000º	-87.829º	2.41 %	Inflection point of the underdamped system ($\zeta \approx 0.155$).
Magnitude at 10 kHz (f_C)	-7.330 dB	-7.340 dB	0.13 %	Determine the gain that the attenuator should provide at 10 kHz.
Phase at 10 kHz (f_C)	-148.820º	-158.531º	6.52 %	Actual phase shift of the plant relative to the desired crossover frequency.
Zero by *ESR* (f_z)	28,952.42 Hz	≈ 28.950 Hz	< 0.01 %	Frequency at which the *ESR* breaks the decline to -20 dB/decade.

4. **Physical Analysis of the Results and Discrepancies**

- *Low-Frequency Gain (10 Hz):* The measurement of 27.518 dB matches the equation exactly. The voltage divider formed by the copper resistance of the inductor (r_L = 25 mΩ) and the output load (R_L = 2.5 Ω) reduces the average DC voltage from 24.0 V to 23.762 V.
- *LC Filter Resonance (1.30 kHz):* The *LC* filter has a measured quality factor of Q = 3.22, producing a peak of +37.675 dB. This underdamped behavior causes the phase to drop rapidly to -90° at resonance.
- *Target Crossover Frequency (10 kHz):* At a frequency of 10.0 kHz (f_{sw} / 10), the circuit attenuates the signal by -7.340 dB (a linear factor of ≈ 0.4295) and exhibits a severe phase shift of -158.531°.

Explanation of the phase discrepancy at 10 kHz (6.52%): In the simplified first-order analytical derivation, the high-frequency cross-admittances of the sources dependent on the Vorperian model are omitted. The SPICE simulator solves the complete circuit matrix, accurately capturing the additional phase delay accumulated before the zero region due to *ESR*.

5. **Required Specifications for the Type II Compensator**

To stabilize the plant in a closed-loop system while ensuring a target phase margin of Φ ≥ 60° and a minimum settling time without oscillations, the analog compensation network based on an operational amplifier must meet two strict requirements at the crossover frequency f_C = 10.0 kHz:

5.1. **Profit Condition**

The analog compensator $A_v(s)$ must provide an exact gain that cancels out the system attenuation at 10 kHz, forcing the crossover of the total loop $T(s) = A_v(s) \cdot G_{vd}(s)$ to 0 dB:

$$|A_v(j2\pi{\cdot}10\ kHz)|_{dB} = -|G_{vd}(j2\pi{\cdot}10\ kHz)|_{dB} = -(-7.340\ dB) = +7.340\ dB$$

In linear magnitude:

$$A_{v.fc} = 10^{\frac{7.340}{20}} \approx 2.3281$$

5.2. **Phase Condition**

The closed-loop phase margin Φ is expressed as:

$$\Phi_m = 180^{\circ} + \angle G_{vd}(f_C) + \phi_{boost} \geq 60^{\circ}$$

Substituting the actual phase shift measured in Ltspice (angle $G_{vd}(f_C)$ = -158.531º):

$$60^{\circ} = 180^{\circ} - 158.531^{\circ} + \phi_{boost} \Rightarrow 21.469^{\circ} + \phi_{boost} = 60^{\circ} \Rightarrow \phi_{boost.theoretical} = 38.531^{\circ}$$

By adding a safety margin of +10.0° to account for the phase degradation caused by the finite gain-bandwidth product (*GBW*) of the actual operational amplifier and the tolerances of the discrete passive components (0.1% for resistors and 1% for polypropylene capacitors), we set the design requirement at:

$$\phi_{boost.design} \approx +48.53^{\circ}\ (at\ 10.0\ kHz)$$

6. **References and Bibliography**

- *Vorperian, V. (1990).* Simplified analysis of PWM converters using a model of the PWM switch. Part I: Continuous conduction mode. IEEE Transactions on Aerospace and Electronic Systems, 26(3), 490–496.

- *Erickson, R. W., & Maksimović, D. (2001).* Fundamentals of Power Electronics (2nd ed.). Springer Science & Business Media.

- *Basso, C. (2012).* Designing Control Loops for Linear and Switching Power Supplies: A Tutorial Guide. Artech House.

- *Analog Devices / Linear Technology. (2010).* LTspice IV User Guide: Small-Signal AC Analysis and .MEASURE Directives. Analog Devices Technical Documentation.